

G06: Towards a Unified World Height System

Geoid of Nepal from Airborne Gravity Survey

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Abstract

An airborne gravity survey of Nepal was carried out December 2010 in a cooperation between DTU-Space, Nepal Survey Department, and NGA, USA. The entire country was flown with survey lines spaced 6 nm with a Beech King Air aircraft, with a varying flight altitude from 4 to 10 km. The survey operations were a major challenge due to excessive jet streams at altitude as well as occasional excessive mountain waves. Despite the large 400 mGal + range of gravity anomaly changes from the Indian plains to the Tibetan Plateau, results appear accurate to few mGal, with proper evaluation from cross-overs complicated by the high varying flight altitudes. Using a downward continuation scheme based on blocked least-squares collocation, a new geoid of Nepal was computed by Fourier methods. The new geoid shows large changes to EGM08, illustrating the impact of the new data. The new geoid is compared to limited GPS-levelling data as well as recent GPS-heights of Mt. Everest, and the new data also provide an independent validation of GOCE gravity field models at the local ~100 km resolution scale.

Keywords

Airborne gravimetry • Geoid • Nepal

1 Introduction

The Danish National Space Institute, Technical University of Denmark (DTU-Space), in cooperation with the Survey Department, Nepalese Ministry of Land Reform and Management, with primary economic support from the US National Geospatial-Intelligence Agency (NGA), carried out an airborne survey of all of the Kingdom of Nepal in December 2010, with the primary goals of the airborne survey to provide data for a new national geoid model, in support of GPS surveying and the national geodetic infrastructure, as

well as provide gravity information for future global gravity field (EGM08 successor) models.

Additional purposes of the project were to provide training for Nepalese geodesists in physical geodesy, as well as to make an improved estimate of the geoid at Mt. Everest, allowing an independent height determination by GPS. Finally it also turned out that the new airborne data provided an independent validation of GOCE and EGM08 in this most mountainous part of the earth.

The airborne gravity surveys (Fig. 1) were carried out using a Beech King Air aircraft, provided by the Danish company COWI. A total of 57 flight-hours were flown at a track spacing of six nautical miles in a total of 13 flight days December 4–17, 2010, excluding transfer flights to and from Denmark. The coverage of the border regions of the country was not possible, since the survey aircraft was not allowed to enter the airspace of China or India (Fig. 2).

The survey was flown at varying elevations, because of the obvious constraints of the terrain, with the highest peaks of

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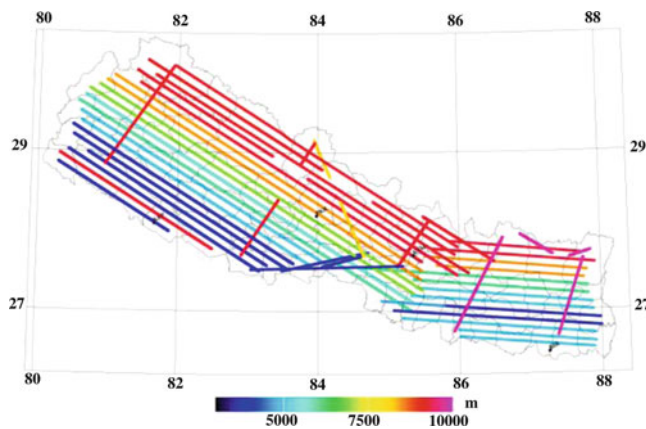


Fig. 1 Flight tracks of the Nepal airborne gravity survey. Colours show the flight altitude, ranging from 3,700 to 9,850 m



Fig. 2 COWI Beech King Air aircraft used for the 2010 Nepal survey flights measurements

the Himalaya in the north along the Chinese border, but also in the central Annapurna range, cf. Fig. 3. Overall flight conditions were extremely challenging over the high mountains, with jet streams occasionally in excess of more than 100 knots wind speed providing excessive turbulence and drafts. Some parts of the lines, in spite of reflight attempts, did not provide useful data due to the persistent wind effects; these areas in the central north (Annapurna and Mustang Valley regions) are evident in Fig. 1 showing only the parts of the lines with processed gravity data; in these regions must therefore be augmented with surface gravimetry data.

All gravity measurements were made using a Lacoste and Romberg S-type gravimeter running the Ultrasys control system, with a number of GPS receivers onboard the aircraft and on ground providing the necessary kinematic positioning. The processing of the airborne gravity data was done following the principles outlined in Olesen (2003), with a typical full-width filter length of about 180 s, corresponding

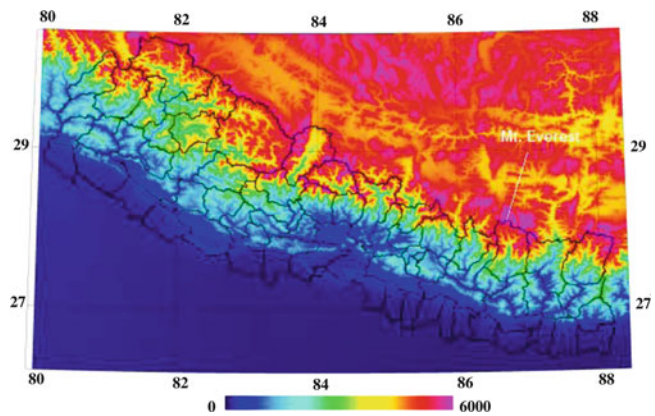


Fig. 3 Terrain model of Nepal from SRTM data

to a resolution of around 6–8 km, depending on wind speed. In addition a Russian Chekan-AM gravimeter was flown for the first time alongside the LCR gravimeter, with the special goal to augment the LCR gravity data in the expected turbulent conditions (the results of the Chekan gravimeter will be presented elsewhere).

The airborne gravity reference point at Kathmandu airport was tied to recent NOAA absolute gravity measurement sites at the Survey Department and the fundamental geodetic observatory at Nagarkot, and the airborne gravity thus well-controlled in absolute level. The actual gravity values at altitude differ by 1,000's of mGal from the reference values due to the height changes and Eötvös corrections, and obtaining bias-free results is challenging, since the scale factors of the used gravimeters might not be sufficiently accurate.

The r.m.s. accuracy of the airborne free-air gravity anomalies cannot be judged from cross-over adjustments, due to the excessive flight height differences. Performing a collocation estimation of vertical gravity gradients, and applying these gradients to continue the airborne gravity to a common level on a line-by-line basis, indicates an r.m.s. error at 94 line crossings around 3.3 mGal for the terrain-reduced data, a highly satisfactory result given the very challenging flight conditions in Nepal. For comparison the DTU airborne gravity results in more benign regions typically give r.m.s. accuracies at the 1.5–2 mGal level. Details of the downward continuation process used in this study, as applied for the geoid determination, are outlined in the next section.

2 Downward Continuation of Airborne Data and Merging with Surface Data

The harmonic downward continuation of airborne gravity data is an inherently unstable operation, making the collocation approach or stabilized Fourier methods the method of choice for treating this problem (Forsberg 2002).

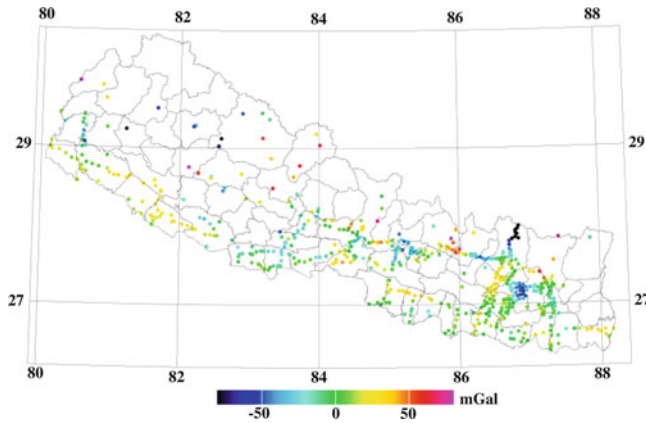


Fig. 4 Surface gravimetry in Nepal, as provided by Survey Department and Italian sources (Poretti et al., personal communication)

Downward continuation is essentially a high-pass filtering operation, which will amplify short-wavelength noise in the airborne data. For geoid determination, this is offset by the gravity to geoid low-pass filtering operation; therefore, airborne gravity is well-suited for geoid determination. Use of airborne gravity for geoid boundary value problems has been investigated among others by Li (2000); for more general studies on the downward continuation problem see, e.g., Keller and Hirsch (1992).

Surface gravimetry will strengthen the downward continuation process. For Nepal, only a very sparse surface gravity data set exists, limited to the main valleys. All surface data were merged and checked for large gross-errors prior to the use (but many minor errors in the older data are still evident). Figure 4 shows the terrain- and EGM-reduced surface gravity anomaly data used in the preliminary geoid computation of this paper.

Because of the mix of airborne and surface data, and the varying flight altitudes, we use the method of least squares collocation for the downward continuation. We thus generate a unified 3' surface free-air gravity grid at the surface of the terrain from a combination of all data (after reduction from terrain and reference field), to be used for the subsequent geoid computations.

The least squares-collocation downward continuation estimates the signal “s” at a ground grid point from a vector “x” containing all available surface and airborne data by (Heiskanen and Moritz 1967):

$$\hat{s} = C_{sx}[C_{xx} + D]^{-1} \quad (1)$$

Covariances C_{xx} and C_{sx} are taken from a full, self-consistent spatial covariance model, and D is the (diagonal) noise matrix. Because the gravity field of the earth is known to follow Kaula’s rule, it is important to select covariance models which have an implied power spectral density decay

in accordance with this. An example of such a self-consistent covariance model is the spherical earth Tscherning–Rapp model (Tscherning and Rapp 1974), and the analogous planar logarithmic covariance model (Forsberg 1987). In this model, gravity covariance between gravity anomalies at two altitudes is of form:

$$C(\Delta g^{h_1}, \Delta g^{h_2}) = - \sum_k \alpha_k \log(D_k + \sqrt{s^2 + (D_k + h_1 + h_2)^2}) \quad (2)$$

where α_k are weight factors combining terms relating to depth value terms ($D_k = D + kT$), with the “free parameters” D and T taking the role analogous to the Bjerhammar sphere depth of spherical collocation and a “compensating depth” attenuation factor.

All processing has been done using the remove-restore method, where the gravity anomalies have been split into three terms

$$\Delta g = \Delta g_1 + \Delta g_2 + \Delta g_3 \quad (3)$$

where the first term is a spherical harmonic reference field, the second term the terrain effects, and the third part the residual gravity.

For the spherical harmonic reference field, a composite model of the EGM08 (Pavlis et al. 2008) and a second generation GOCE model (Pail et al. 2010) has been used. The EGM08 and GOCE models have been merged, so that the GOCE coefficients are in a central mid-wavelength resolution band, by fully substituting EGM08 coefficients with GOCE in the band 90–160, and linearly blending GOCE and EGM08 in the border bands 80–90 and 160–170.

In this way the long-wavelength GRACE information in EGM08 is kept, and the new GOCE information used at the wavelength scales relevant to Nepal, and the information from terrestrial data in EGM08 kept above the GOCE resolution. Because the terrestrial data sources behind EGM08 are unknown (and based upon 15' mean values in the region), the composite reference model is only used to maximal spherical harmonic degree 720.

The composite model, termed EGM in the sequel, give the geopotential V as a three-dimensional expansion:

$$V(r, \varphi, \lambda) = \frac{GM}{R} \sum_{n=0}^{720} \left(\frac{R}{r}\right)^n \times \sum_{m=0}^n (C_{nm} \cos m\lambda + S_{nm} \sin m\lambda) P_{nm}(\sin \varphi) \quad (4)$$

To properly take into account the height dependency of the spherical harmonics, a “sandwich grid” has been used. Here gravity anomalies Δg_1 (and height anomalies ζ_1) are

Table 1 Statistics of the gravity data reductions

Data	Mean	Std.dev.
Airborne gravity data	−13.8	119.8
Airborne minus ref. field	2.6	31.6
Airborne minus ref. and RTM	5.1	26.4
Surface gravimetry data	−87.2	63.9
Surface minus ref. and RTM	1.1	28.1

Unit mGal

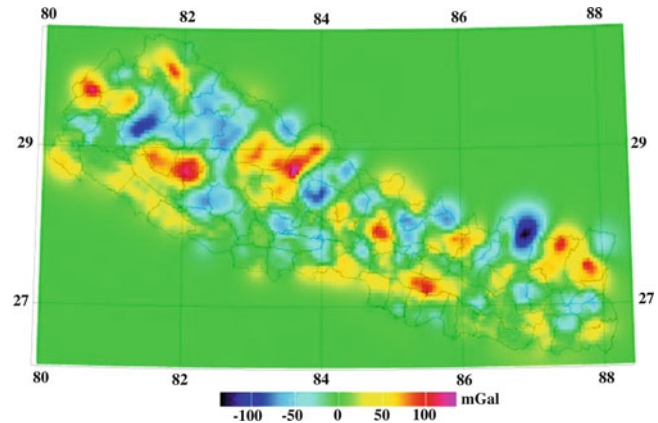
evaluated at grids at two elevations (0 and 9 km), from which both airborne and surface data are interpolated linearly with respect to both latitude, longitude and height. This takes into account the height dependency of the spherical harmonic expansion.

The use of gravity terrain effects in the downward continuation process has significant advantages: it will—if properly applied—reduce the short-wavelength information in the gravity data, and thus make the problem more stable; it will also diminish the topographic aliasing, especially important for surface gravity data, which typically do not have a random distribution relative to the topography (usually there is a relative abundance of gravity data collected in valleys).

In Nepal SRTM elevation data, averaged to 15'' resolution, have been used to compute RTM terrain effects Δg_2 on all surface and airborne data. The RTM terrain effects (Forsberg 1984) takes into account topography relative to a mean height surface. This surface is produced from the SRTM data by low-pass filtering. We used a Gaussian filter with a resolution of 0.5° to produce the RTM mean height surface. The RTM terrain effects are computed by prism integration using the TC program (Forsberg 1984), assuming a constant density of 2.67 g cm^{−3} for all topography. For airborne gravity data one important aspect must be taken into account: because airborne gravity data are inherently along-track *filtered*, the terrain effects computed at altitude must be filtered by the same filter algorithm as used in the airborne processing.

Table 1 shows the results for the reduction of the airborne and surface data for the EGM reference field and the RTM terrain effects. It is seen that the airborne data are apparent somewhat biased, but this might also come from limited size of the Nepal area (less bias has later been found with third generation GOCE models), but could potentially also come from the earlier-mentioned possible errors in the manufacturer-provided gravimeter scale factors.

For the final collocation run with the planar logarithmic model (2), D and T values were fitted to the gravity data (D = 10 and T = 35 km seemed a good fit for surface and airborne data). Standard errors were assigned to data as follows: airborne gravity 3 mGal, and surface gravimetry 5 mGal (to limit the influence of residual gross errors in the surface data, and give more weight to the more homogenous airborne

**Fig. 5** Downward continued reduced gravity grid at surface, using EGM08 + GOCE ref. Colour range −135 to 135 mGal

data). The collocation downward continuation was made in one solution, predicting a surface free-air anomaly grid of 3' resolution, shown in Fig. 5. This grid represents the influence of the local data, and shows that the EGM08 + GOCE reference model catch the long-wavelength gravity variations well.

3 Gravimetric Geoid Computation

Using the reduced gravity data, gravimetric geoid models were computed by spherical FFT methods.

In the multi-band FFT method used in this study (Forsberg and Sideris 1993), the fundamental Stokes' formula is transformed into a spherical convolution in latitude and longitude (φ, λ) for a given reference parallel φ_{ref} , and by utilization of a number of bands a virtually exact convolution expression may be obtained by a suitable linear combination of the bands. For each band the convolution expressions are evaluated by

$$\begin{aligned} \zeta_3 &= S_{\text{ref}}(\Delta\varphi, \Delta\lambda) * [(\Delta g_3 + g_1^c) \sin \varphi] \\ &= F^{-1} [F(S_{\text{ref}}) F(\Delta g \sin \varphi)] \end{aligned} \quad (5)$$

where S_{ref} is a (modified) Stokes' kernel function, g_1^c the terrain reduced Molodensky term (Forsberg and Sideris 1989), also computed by FFT, and * and F are the two-dimensional convolution and Fourier transform, respectively.

In the actual implementation of the method, the data are gridded by least-squares collocation, and a 100 % zero padding is used to limit the periodicity errors of FFT.

A modified Wong–Gore Stokes kernel is used, where only spherical harmonics above degree 80 are allowed to modify the underlying EGM08 GRACE-based reference field. To avoid edge effects, the Wong–Gore kernel modification is

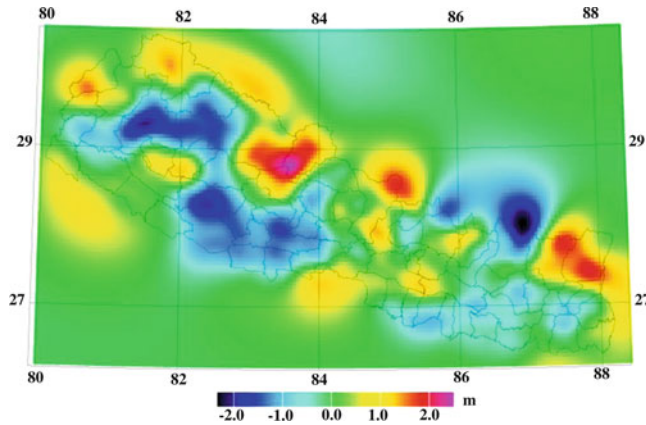


Fig. 6 Residual geoid effects from FFT of the airborne data (Fig. 5). This plot shows the effects of the airborne and surface gravity data, relative to GOCE/EGM08. Range -2.2 to 2.8 m

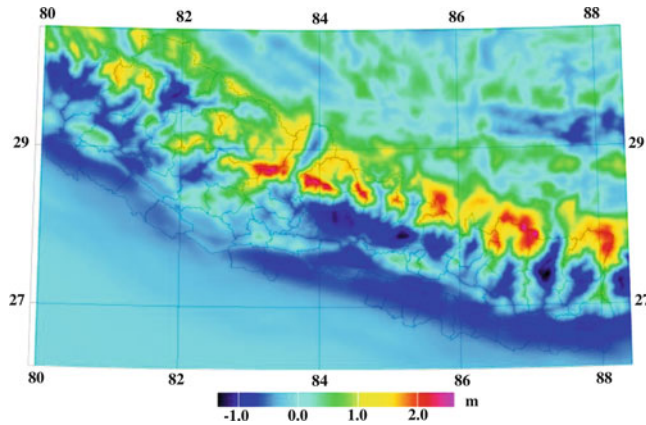


Fig. 7 RTM terrain effects on the geoid. Scale -1.2 to 2.8 m

linearly blended from degree 80 to 90, for details see Forsberg and Featherstone (1998).

After the FFT transformation to the (residual) quasigeoid (Fig. 6), the geoid terrain effects (Fig. 7) and the spherical harmonic terrain effects, computed at the level of the topography, are restored to give the final quasigeoid, shown in Fig. 8.

To obtain the final classical geoid model, the quasigeoid is converted to geoid N to first order by use of the equation (Heiskanen and Moritz 1967):

$$\zeta - N \approx -\frac{\Delta g_B}{\gamma_0} H \quad (6)$$

where Δg_B is the Bouguer anomaly, H the height, and γ_0 normal gravity. The geoid–quasigeoid separation is shown in Fig. 9. It is clear that the magnitude of the terms are very large, and higher order approximations are needed for e.g. the most precise geoid determinations, not taken into account in this preliminary geoid computation.

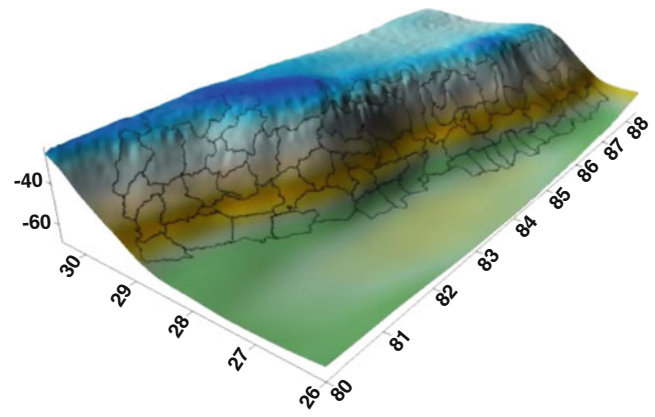


Fig. 8 Quasigeoid of Nepal. Range N–S is more than 40 m

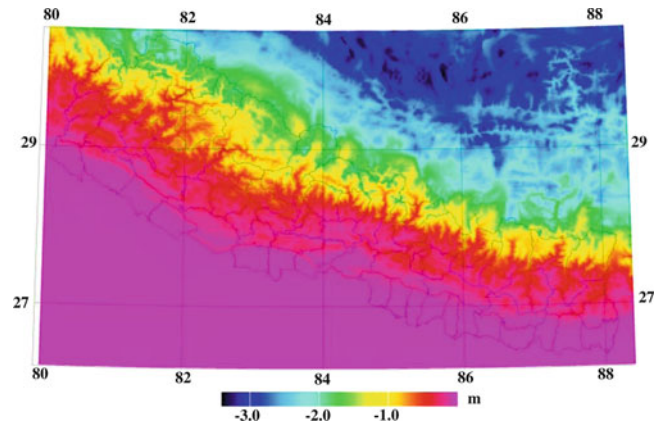


Fig. 9 Quasigeoid minus classical geoid. Scale -3.5 to 0 m

4 GOCE Comparison to the Airborne Data

The new airborne gravity data provide a validation opportunity for a check of local-scale gravity anomalies at the ~ 100 km resolution. Figure 10 shows the GOCE gravity anomalies (maximal degree 180) over Nepal, as well as the airborne data. It is apparent that the local anomalies seen by GOCE are also evident in the airborne data. The GOCE anomalies are quite different to EGM08, as illustrated in Fig. 11 showing the geoid difference between EGM08 and EGM08/GOCE combination field used for the local geoid computation, both to degree 720.

5 Geoid Comparison to GPS and Levelling

Limited reliable GPS-levelling derived geoid heights exist in Nepal. Table 2 shows the comparison to eight reasonable precise data in the Kathmandu Valley. It is seen that a significant

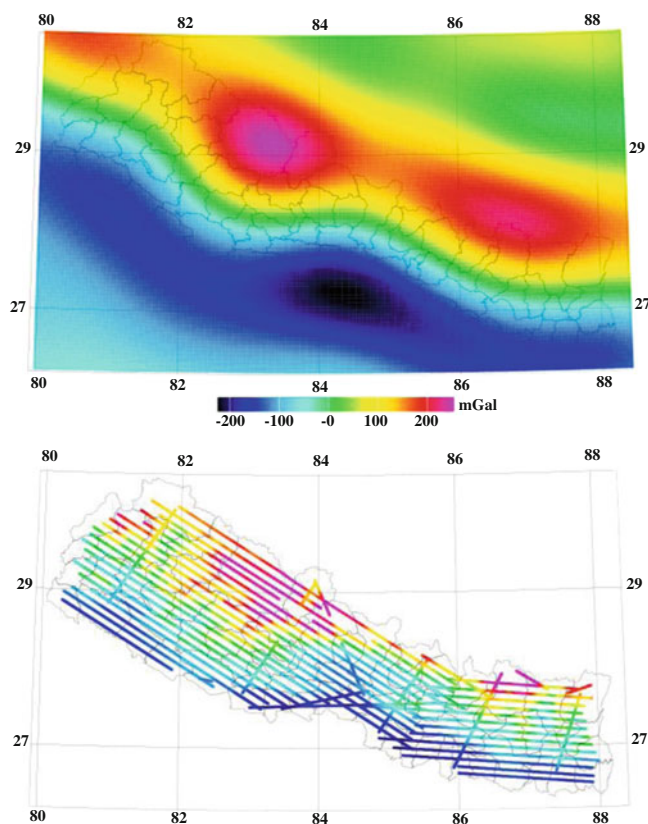


Fig. 10 GOCE gravity anomalies (*top*) and airborne gravity anomalies at altitude (*lower*). Scale -220 to 240 mGal

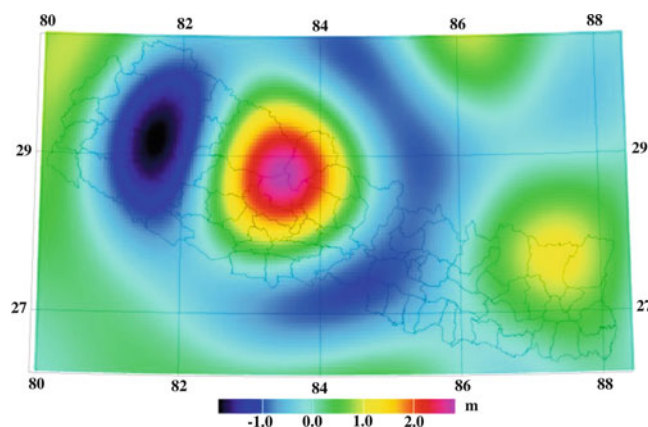


Fig. 11 Geoid difference between GOCE and the composite GOCE + EGM08 combination model. Scale -1.8 to 2.8 m

geoid improvement has been obtained relative to the EGM08 geoid, and also that the official GPS heights in Nepal are not conventional ellipsoidal heights, but relative ellipsoidal heights relative to the fundamental Nagarkot datum point. Therefore a recomputation of the national GPS heights is needed before any fitting of the new geoid to GPS-levelling data.

Table 2 Geoid comparison, GPS-levelling Kathmandu Valley

Unit (m)	Mean	Std.dev.
New geoid, using all data	18.85	0.11
New geoid, airborne data only	18.95	0.11
EGM08 geoid (degree 2,160)	18.51	0.30
GOCE geoid (degree 160)	19.16	0.42

For the height of Mt. Everest, the geoid values at the mountain are highly dependent of the correction details of the computations, especially the geoid–quasigeoid separation. The computed geoid heights of Mt. Everest confirms the official Survey Department height of $8,848$ m within a range of ± 1 m, based on an ellipsoid height of the snow summit of $8,821.40 \pm 0.03$ m (Chen et al. 2006). Improved geoid computations using recent updated GOCE data, the Chekan data, and more surface gravimetry—hopefully also on the Chinese side of the border—are needed to obtain the best height of the summit.

6 Conclusions

A new airborne gravity survey has successfully covered all of Nepal in 2010. The overall survey accuracy, without using any internal cross-over adjustment, is estimated at the 3.3 mGal r.m.s. level, with some minor errors in bias possible due to uncertainties in gravimeter scale factors.

We have in the paper outlined the procedure of downward continuation by least-squares collocation and geoid computation by spherical FFT and GOCE. The accuracy of the computed geoid is likely at the 10 – 20 cm in the interior of Nepal (based on formal least squares error estimates), but higher near the borders due to lack of data in China and India.

The outlined geoid determination was preliminary, and awaits the final airborne gravity sets, as well as use of enhanced geoid determination theory and the most recent GOCE models.

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GOCE and the Geoid in South America

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Abstract

The global geopotential models GO_CONS_GCF_2_DIR_R2, GOCO02S and EIGEN06C based on GOCE data, and EGM08, represent a new important contribution to medium and long wavelength components knowledge of the gravitational field. These models have impacts on the computation of the respective wavelength components of the geoid. They have been used as reference fields in the modified Stokes integral, generating different geoid models. The terrestrial gravity data in South America have been updated with the most recent measurements in Argentina, Brazil, Ecuador and Paraguay. The short wavelength components were estimated via FFT using Featherstone modified kernel. The complete Bouguer and Helmert gravity anomalies have been derived through the geoid modelling package SHGEO by University of New Brunswick, Canada. The GGMs and the geoid models have been evaluated using GPS observations on Bench Marks of the spirit levelling network (GPS/BM). The height anomaly derived from EGM08 (degree 2190 and order 2159) has also been checked out.

Keywords

Geoid modeling • GOCE • GPS • GRACE • Height

1 Introduction

A new version of the geoid model for South America (GEOID2010) was computed in an area limited by 15° N and 57° S in latitude and 30° W and 95° W in longitude in the Laboratory of Topography and Geodesy, University of São Paulo. The model was computed using EGM08 (Pavlis et al. 2008) up to degree and order 150 as a reference

field. The reduced Helmert mean gravity anomalies were estimated in blocks of 5' for continental areas. The oceanic region was completed with the mean free-air anomalies derived from a satellite altimetry model by the Danish National Space Center called DNSC08 (Andersen et al. 2009). The digital terrain model SAM3s_v2, with a grid size of 3" × 3" (~90 m × 90 m), was chosen for computing the new geoid. The complete Bouguer, Helmert gravity anomalies and primary indirect topographical effect have been derived through the Canadian package SHGEO. The short wavelength component (residual co-geoid) was estimated via FFT and the modified Stokes kernel proposed by Featherstone (2003) was used. The Brazilian territory of this model, called MAPGEO2010, has been adopted as the official geoid model in Brazil since 2010 by Brazilian Institute of Geography and Statistics (IBGE).

The other geoid models were computed with the same procedure as GEOID2010, using as reference spheroid the satellite-only models GO_CONS_GCF_2_DIR_R2

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